

# Chromosome-level de novo genome assembly and whole-genome resequencing of the threatened species *Acanthochlamys bracteata* (Velloziaceae) provide insights into alpine plant divergence in a biodiversity hotspot

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## Abstract

The Hengduan Mountains region is an important hotspot of alpine plant diversity and endemism. *Acanthochlamys bracteata* is a species of a threatened monotypic genus endemic to the Hengduan Mountains. In this study, we present a high-quality, chromosome-level reference genome for *A. bracteata*, constructed using long reads, short reads and Hi-C technology. We characterized its genetic diversity, population structure, demographic history and gene flow by resequencing individuals collected across its distribution. Comparative genomics analyses based on sequence information from single-copy orthologous genes revealed that *A. bracteata* and *Dioscorea rotundata* diverged ~104.5 million years ago. Whole-genome resequencing based on population genetic analysis revealed that the division of the 14 populations into 10 distinct clusters reflected geographical divergence, and three separate high levels of gene flow occurred sequentially between isolated populations of the Hengduan Mountains, a finding which is consistent with the turnover between ice ages and interglacial periods. Our findings indicate that Quaternary climatic changes played an important role in shaping the genetic structure and demographic trajectories of *A. bracteata*, and provide critical insights into the genetic status and evolutionary history of this poorly understood species, and possibly other alpine plants with a similar distribution. This study demonstrates the usefulness of population genomics for evaluating the effects of past climatic changes and identifying conservation units for the conservation and management of threatened species. Our high-quality genome represents a valuable resource for future studies of the underlying molecular mechanisms of adaptive evolution and provides insight for further comparative genomic analysis with other Velloziaceae species.

## KEYWORDS

*Acanthochlamys bracteata*, gene flow, Hi-C, population genomics, threatened species

## 1 | INTRODUCTION

The Hengduan Mountains (HDM) region, one of the 35 recognized diversity hotspots worldwide (Williams et al., 2011), is characterized by an extreme diversity of plant species. It is estimated that 12,800 species exist in an area of about 500,000 km<sup>2</sup> (Sun et al., 2017). The HDM are one of the most ecologically diverse areas in the Qinghai-Tibet Plateau and adjacent regions, with at least 3300 endemic plant species existing in this region. Thus, the unique, abundant and diverse flora of the HDM, resulting from a high degree of topographical diversity and climatic oscillations, have made this area one of the most important regions for studying the population differentiation and speciation of plants. Previous studies have focused primarily on tree (Du et al., 2017; Liu et al., 2013; Li, Milne, et al., 2020; Xu et al., 2010) and shrub species (Chen et al., 2019; Gao et al., 2019; He et al., 2021; Jian et al., 2015), while only a few studies have focused on herbaceous plant species (Ren et al., 2020; Wang et al., 2010). In the HDM region, the study of distribution-restricted and threatened species is particularly important for the conservation of species diversity in the region.

Worldwide declines in species numbers have been the subject of considerable research over several decades. However, in many species there is also a loss of intraspecific variation, producing a hidden biodiversity crisis (Des Roches et al., 2021). The identification of conservation units (CUs) is important for the conservation and management of threatened species, providing managers and policy-makers with a clear understanding of the boundaries of population units (Funk et al., 2012; Li, Milne, et al., 2020; Li, Wang, et al., 2020). A comprehensive understanding of population structure, genetic differentiation and demographic history is crucial for an understanding of the evolutionary history of a species, and the implications of these issues for the conservation of the organisms (Wang et al., 2019). Genetic markers such as microsatellite loci, chloroplast DNA and mitochondrial DNA may be inadequate for determining the population genetic structure of species with complex demographic histories. Population genomics based on resequencing technology can identify numerous single-nucleotide polymorphisms (SNPs), the analysis of which has the potential to provide a deeper understanding of adaptive patterns and CUs among different distributions (Funk et al., 2012).

The monotypic *Acanthochlamys* P. C. Kao is a genus of herbaceous plants that is sometimes placed in its own family, Acanthochlamydaceae (Kao, 1998; Kao & Kubitzki, 1998), but is generally considered to be a basal member of the predominantly Southern Hemisphere family Velloziaceae (Mello-Silva et al., 2011; Salatino et al., 2001). *Acanthochlamys bracteata* P. C. Kao, the only known species of the genus, was a member of the flora to the south of the Tethys or in northern Gondwana (Sun et al., 2017), but is now a relic native to western Sichuan and eastern Tibet, mainly around the Jinsha River and its tributaries. Morphologically, it is a plant with a clumping stem, recognized as meadow plant in the HDM, with populations distributed at elevations of around 2700–3800 m. This species has many particular anatomical, embryological,

cytological and palynological special characteristics (Kao, 1998), and is now classified as “Vulnerable” by the International Union for Conservation of Nature (IUCN) (Chadburn, 2017), and as a “Second-Class National Key Protected Wild Plant” by the China National Forestry Administration and Department of Agriculture in 1999.

Approaches using whole genome and Hi-C sequencing, or transcriptomics, are opening new research pathways. We have produced a high-quality genome sequence of *A. bracteata*, which was used as a reference for a population genomics study of 110 individuals from 14 populations. The aims of our study were to: (i) produce a high-quality genome assembly of *A. bracteata* as an important genomic resource and reference for future studies; (ii) evaluate the genetic diversity, population structure, demographic history and gene flow of *A. bracteata* and understand the driving factors involved; and (iii) use our data to assess the conservation implications and make recommendations for the conservation of this threatened species.

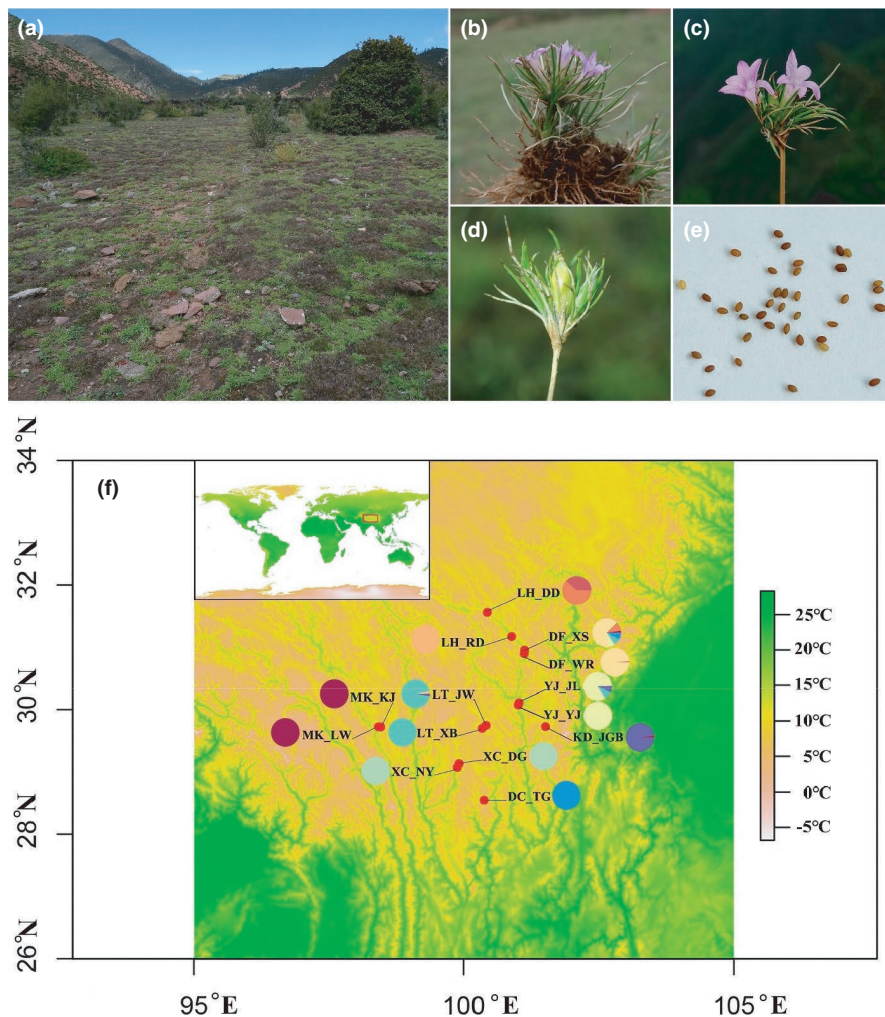
## 2 | MATERIALS AND METHODS

### 2.1 | Sample collection

*Acanthochlamys bracteata* (Figure 1) is restricted to the HDM in southwest China. The sampled individuals were from Lawu village, Rumei Town, Mangkang county, Tibet, China (98.415098°E, 29.721676°N), growing in open scrub or meadows at an elevation of 3450 m. Fresh leaves were immediately frozen in liquid nitrogen and stored at –80°C until DNA extraction. Genomic DNA was extracted from the leaf tissues following the CTAB protocols (Doyle & Doyle, 1987). Leaves, stems, flowers and buds were collected for RNA extraction using Biomarker Plant Total RNA Isolation Kits (rich in polysaccharides and polyphenolics). Leaf samples for all library preparations are from different individuals from the same cluster.

### 2.2 | Library construction and genome sequencing

For Nanopore sequencing, high-quality genomic DNA was extracted and sequenced following the instructions for Ligation Sequencing Kits (SQK-LSK109; Nanopore). The DNA was purified, and high-quality data were obtained. The DNA was sheared in a SonicMan microplate sonicator (Brooks Automation), to produce 50 µg of high-quality genomic DNA with a median fragment size of ~20 kb, which was further enriched and purified using Agencourt AMPure XP beads. Damaged DNA and ends were enzymatically repaired using the NEB Next End Repair/dA-Tailing Module (NEB), and the quantity of DNA was assessed using a Qubit 2.0 Fluorometer (Thermo Fisher). Finally, a 20-kb library was added to FLO-PRO002 flowcells and run on Nanopore PromethION platform at Biomarker Technology Company. For Illumina PCR-free sequencing, paired-end libraries with insert sizes of 350 bp were constructed, and the Illumina short reads which were first used to estimate genome size were used following the Illumina company protocol and sequenced on an Illumina



**FIGURE 1** Morphological characters of *Acanthochlamys bracteata*, and geographical distribution of population samples. (a) Habitat. (b) Habit. (c) Flower. (d) Fruit. (e) Seeds. (f) Sample locations of the 14 populations in the HDM based on bioclimatic factors (mean temperature of warmest quarter). Pie charts at sampling locations indicate the distribution of genetic groups identified by a STRUCTURE analysis of the population

Hiseq 2500 platform. For transcriptome sequencing, a total of 3  $\mu$ g genomic RNA from different tissues, such as leaves, stems and flowers, was used as input material for the RNA sample preparations. Sequencing libraries were generated using NEBNext Ultra RNA Library Prep Kits for Illumina (NEB) following the manufacturer's protocol. Index codes were added to each sample. The library was sequenced on an Illumina Hiseq 2500 platform and paired-end reads were generated.

### 2.3 | Dovetail Hi-C library preparation and sequencing

Dovetail Hi-C library preparation and sequencing were carried out by Dovetail Genomes. A Dovetail Hi-C library was prepared as previously described (Lieberman-Aiden et al., 2009). For each library, chromatin was fixed in place with formaldehyde in the nucleus, and the fixed chromatin was then extracted and digested with *DpnII*. The 5' overhangs were filled in with biotinylated nucleotides, and the free blunt ends were then ligated. After ligation, the cross-links were reversed, and the DNA was purified from the protein. The purified DNA was treated to remove biotin that was not internal to the

ligated fragments. The DNA was then sheared to a mean fragment size of ~350 bp. The DNA with biotin was captured using adsorption to affinity beads, and the whole library preparation was completed by end repair, ligation with A-junctions, and PCR amplification and purification of the DNA fragments, in strict accordance with the operation procedure. Qubit 2.0 was then used for initial quantification, and the libraries were diluted to 1 ng  $\mu$ l<sup>-1</sup>, followed by the detection of the insert size of the libraries using an Agilent 2100. When the insert size was as expected, the effective concentration of the libraries was accurately determined using quantitative PCR to ensure the quality of the libraries. After passing the library check, the constructed libraries were sequenced using an Illumina Hiseq 2500 platform. The filtered Hi-C reads were aligned to the initial draft genome by BWA version 0.7.17 (Li & Durbin, 2009) software which was integrated into JUICER version 1.6.2 (Durand, Robinson, et al., 2016; Durand, Shamim, et al., 2016) software. Only uniquely mapped and valid paired-end reads were used for assembly by 3D-DNA (Dudchenko et al., 2017). JUICEBOX version 1.13.01 (Durand, Robinson, et al., 2016; Durand, Shamim, et al., 2016) was used to manually order the scaffolds to get the final chromosome assembly. Contact maps were plotted with HICEXPLORER version 3.0 (Wolff et al., 2020).

## 2.4 | De novo genome assembly

Before assembly, 17-mer analysis based on the PCR-free Illumina short reads led to an estimated genome size of ~245 Mb (Figure S1). A draft genome of *A. bracteata* was assembled using the program WTBG2.5 (Ruan & Li, 2020) with total clean Nanopore long reads as input, 40 thresholds (CPU cores) and other default parameters used for program settings. Then, RACON version 1.4.3 (Vaser et al., 2017) was used to calibrate the draft genome with Nanopore reads with three iterations. Finally, Illumina short reads were mapped back to the contigs/scaffolds and further polished using NEXTPOLISH version 1.3.1 (Hu et al., 2020) for three iterations.

## 2.5 | Assessment of assembly quality

To assess the quality of the assembly, the completeness of the final genome assembly was evaluated using Benchmarking Universal Single-copy Orthologues (BUSCO) (Simão et al., 2015). The BUSCO pipeline version 3.1.0 was used to test for the presence and completeness of orthologues using the Embryophyta\_odb10 database, which contained 1614 single-copy orthologous genes.

To further curate the assembly, we mapped the Illumina reads back to the draft assembly using BWA version 0.7.17 with default parameters. All the scaffolds were then searched against the Uniprot database (<https://www.uniprot.org/>), and the results of the protein comparison scaffold were obtained using DIAMOND version 0.9.24 with settings: --outfmt 6 --sensitive --max-target-seqs 1 --evaluate 1e-25. The alignments and database scan were used to as input to contamination screening using BLOOTOOLS version 1.1.1 (Kumar et al., 2013) by evaluating coverage, GC content and sequence similarity against the Uniprot database of each sequence.

## 2.6 | Repeat annotation, gene prediction and functional annotation

To generate a repeat library, the de novo prediction of repetitive sequences in the unannotated genome assemblies was conducted, based on structure prediction using LTR FINDER (Xu & Wang, 2007) and on ab initio prediction by REPEATSCOUT version 1.0.5 (Price et al., 2005), respectively. Two repeat libraries were classified using PASTECCLASSIFIER (Hoede et al., 2014) and then merged with the Repbase database (Jurka et al., 2005) as the final database of repeated sequences. Finally, REPEATMASKER 4.1.0 (Tarailo-Graovac & Chen, 2009), with the parameters: -nolow -no\_is -norna -engine wublast, was used to predict the repeat sequences of this genome based on the constructed repeat sequence database.

Protein-coding genes were predicted on the basis of three different strategies, including de novo prediction, homologous species prediction and Unigene prediction. GENSCAN version 1.0 (Burge & Karlin, 1997), AUGUSTUS version 2.4 (Stanke & Waack, 2003), GLIMMERHMM version 3.0.4 (Majoros et al., 2004), GENEID version 1.4

(Alioto et al., 2018) and SNAP (version 2006-07-08) (Korf, 2004) were used for de novo prediction. GEMOMA version 1.3.1 (Keilwagen et al., 2018) was used for prediction based on homologous species. The transcripts were assembled using HISAT version 2.0.4 (parameters: --max-intronlen 20000, --min-intronlen 20) (Kim et al., 2015) and STRINGTIE version 1.2.3 (Pertea et al., 2015) with default parameters based on RNA-sequencing (RNA-seq) data (Pertea et al., 2016), and then TRANSDCODER version 2.0 (<http://transdecoder.github.io/>), GENEMARKS-T version 5.1 (Tang et al., 2015) and PASA version 2.0.2 (parameters: -align\_tools gmap, -maxIntronLen 20000) (Campbell et al., 2006) were used for gene prediction. Finally, EVIDENCEMODELER version 1.1.1 (Haas et al., 2008) was used to integrate the prediction results obtained using the above three methods.

Using the Rfam database (Griffiths-Jones et al., 2005), BLASTN was used for genome-wide alignment to identify microRNAs (miRNAs) and ribosomal RNAs (rRNAs). Transfer RNAs (tRNAs) were predicted using TRNASCAN-SE (Lowe & Eddy, 1997). Pseudogenes were identified as follows: GENBLASTA (She et al., 2009) was used to align predicted homologous genes to the genome, and then GENEWISE (Birney et al., 2004) was used to identify premature and frameshift mutations in the predicted genes, to identify the pseudogenes.

To better understand the gene functions, we searched all predicted protein-coding genes against protein databases, including the NR (Marchler-Bauer et al., 2011), KOG (Koonin et al., 2004), GO (Dimmer et al., 2012), KEGG (Kanehisa & Goto, 2000), TREMBL (Boeckmann et al., 2003) and other functional databases using BLAST version 2.2.31 (e-value cut-off  $1e^{-5}$ ) (Altschul et al., 1990). Then KEGG pathway annotation analysis, KOG functional annotation analysis, GO functional annotation analysis and other gene functional annotation analyses were performed.

## 2.7 | Comparative genomics and phylogenetic analysis

We used ORTHOFINDER version 2.3.12 (Emms & Kelly, 2015) to identify orthologous groups in *Acanthochlamys bracteata* and 10 other species: *Arabidopsis thaliana* (L.) Heynh., *Asparagus officinalis* L., *Dioscorea rotundata* Poir., *Musa acuminata* Colla, *Oryza sativa* L., *Phoenix dactylifera* L., *Phalaenopsis equestris* (Schauer) Rchb. f., *Spirodela intermedia* W. Koch, *Zea mays* L. and *Zostera marina* L. (Table S1). The protein sequences of 737 single-copy orthologous genes were first aligned using MUSCLE version 3.7 (Edgar, 2004) and gaps in the alignment were removed with GBLOCKS version 0.9b (Talavera & Castresana, 2007). Subsequently, a phylogenetic tree was reconstructed using RAXML version 8.2.12 with the GTRGAMMA substitution model (Stamatakis, 2014). Divergence times were estimated by MCMCTree of the PAML4.9j package (Yang, 2007) using the relaxed-clock model. We used two secondary calibration points obtained from the TIMETREE database (Kumar et al., 2017) (<http://www.timetree.org/>) to calibrate our phylogeny: appropriately 50 and 160 million years ago (Ma) for the split time of *Zea mays*-*O. sativa* and *Arabidopsis thaliana*-*M. acuminata*, respectively. Significantly



expanded or contracted gene families and gene family sizes of the most recent common ancestor (MRCA) across the phylogenetic tree ( $p < .05$ ) were calculated using the *CAFE* version 2.1 (De Bie et al., 2006) with the gene birth-death ( $\lambda = 0.003$ ) parameters estimated using the maximum-likelihood method.

## 2.8 | Whole-genome resequencing, SNP calling and filtering

Leaf materials of 110 individuals were collected from 14 wild populations in localities across the distribution area of *A. bracteata* (Figure 1; Table S2), including the one used to produce the assembled reference genome sequence as described above (*THP-SET-1562*, CDBI). The genomes of 110 individuals were resequenced using the Illumina HiSeq X Ten platform. Paired-end reads from each of these individuals were aligned to the assembled *A. bracteata* reference genome using *BWA* version 0.7.17 (<https://github.com/lh3/bwa/releases/tag/v0.7.17>) (Li & Durbin, 2009) with default parameter settings. Aligned reads were processed to remove PCR duplicates, sorted and compressed with *SAMTOOLS* version 1.9 (<https://github.com/samtools/samtools/releases/tag/1.9>). The regions around indels were realigned using *GATK-4.0.11.0* (<https://github.com/broadinstitute/gatk/releases/tag/4.0.11.0>).

The production of high-quality genotypes requires downstream procedures. First, *BCFTOOLS* version 1.9 (<https://github.com/samtools/bcftools/releases/tag/1.9>) was used to retrieve genotypes for variable sites and produce a *vcf* file using the commands “*bcftools mpileup -q 20 -Q 20 -C 50*” and “*bcftools call -f, -m*.” Then *VCFTOOLS* version 0.1.16 (<https://github.com/vcftools/vcftools/releases/tag/v0.1.16>) was used to filter the SNPs, and only SNPs matching the following criteria were retained: (i) minor allele frequencies (MAF)  $\geq 0.01$ ; (ii) mapping quality (MQ)  $\geq 30$ ; (iii) mapping depth (DP)  $\geq 4$ ; and (iv) genotype calls in at least 25% of individuals (missing data  $\leq 0.75$ ).

## 2.9 | Population structure, phylogenetic inference and gene flow analysis

The filtered SNPs were converted to bed file format using *PLINK* version 1.07 (Purcell et al., 2007). Population structure was further assessed by Bayesian clustering in *ADMIXTURE* version 1.3.0 (Alexander et al., 2009). The number of clusters *K* was set to vary from 2 to 14, and the best value was chosen based on the lowest cross-validation error. Finally, the population structure was visualized using the *R* (R Core Team, 2013) statistical software.

To resolve the phylogenetic relationships among the 14 populations, *Xerophyta retinervis* was included in the analysis as an out-group species. The *vcf* file was converted to *fasta* file format using the Python tool *vcf-kit* version 0.2.9 (<https://github.com/AnderSenLab/vcf-kit/releases/tag/0.2.9>). The phylogenetic relationships of the populations were reconstructed using maximum-likelihood

(ML) analysis performed with *IQ-TREE* (Nguyen et al., 2015). Principal component analysis (PCA) was performed using the *R* package *LEA* version 1.4.0 (<https://www.rdocumentation.org/packages/LEA/versions/1.4.0>). To estimate the gene flow between populations of *A. bracteata*, we used *TREEMIX* version 1.13 (Pickrell & Pritchard, 2012) with the 472,000 high-quality SNP dataset with the settings *-se -bootstrap -k 500 -m*, and values for *(-m)* ranging from 1 to 6.

## 2.10 | Demographic analysis using PSMC

To reconstruct the detailed demographic history of *A. bracteata*, we applied the pairwise sequentially Markovian coalescent (PSMC) model (Li & Durbin, 2011) to the de novo genome, with the following parameter settings: *-N25 -t15 -r5 -b -p “4+25 \* 2+4+6.”* The PSMC pattern of parameters, and option *-t*, *-N* and *-r* offer initial parameters for the tools. A bootstrapping approach with 100 replicates was performed, to assess the variation in the inferred  $N_e$  trajectories. Generation times of 2, 3 and 5 years, and a neutral mutation rate estimated to be  $6.4 \times 10^{-9}$  (Lynch, 2010) per site per generation were used to convert the population sizes and scaled time into real sizes and time.

## 2.11 | Ecological niche modelling

To further reveal the patterns of distribution shifts of this species, the potential distribution at present and during the Last Glacial Maximum (LGM, ~20,000 years ago, ka) were predicted using ecological niche modelling (ENM) in *MAXENT* version 3.3.3k (Phillips et al., 2006). We obtained 19 bioclimatic factors: annual mean temperature (Bio1), mean diurnal range (Bio2), isothermality (Bio3), temperature seasonality (Bio4), maximum temperature of the warmest month (Bio5), minimum temperature of the coldest month (Bio6), annual temperature range (Bio7), mean temperature of the wettest quarter (Bio8), mean temperature of the driest quarter (Bio9), mean temperature of the warmest quarter (Bio10), mean temperature of the coldest quarter (Bio11), annual precipitation (Bio12), precipitation of the wettest month (Bio13), precipitation of the driest month (Bio14), precipitation seasonality (Bio15), precipitation of the wettest quarter (Bio16), precipitation of the driest quarter (Bio17), precipitation of the warmest quarter (Bio18) and precipitation of the coldest quarter (Bio19). These data were obtained from Worldclim (<http://worldclim.org/version2>) with spatial resolutions of 30 s. We calculated pairwise Pearson's correlation coefficients (*r*; Dormann et al., 2013) for current climate and altitude data across the distributions of all trees constructed in *ENMTOOLS* version 1.4.3 (Warren et al., 2010). Any factor that had a coefficient  $>0.7$  with two or more other factors was excluded. We generated 110 occurrence records (15 locations from population MK, 17 from LH, 18 from DF, 14 from YJ, 16 from LT, 15 from XC, six from DC and nine from KD) with elevation from our field investigations, which covered most of the known area of occupancy of this species. After removing the pseudoreplication

sites, a total of 33 occurrence records were used in the study. A Community Climate System Model version 3.0 coupled model was used to provide the population with a coupled model framework to conduct climate simulations to obtain predictive variables of the species distribution with a spatial resolution of 30 arc-seconds (Collins et al., 2006). The performance of the models was assessed by comparing their values for the area under the receiver operating characteristic curve (AUC). AUC values range from 0 to 1, with a score of 1 indicating perfect discrimination (Fielding & Bell, 1997).

## 2.12 | Cytological observations

To evaluate the cytological characteristics of the plant, we studied three individuals from one population of *A. bracteata* (voucher: China, Sichuan, Yajiang County, Deng et al., XB2181, CDBI). We obtained seeds from the wild, and stored them at  $-20^{\circ}\text{C}$  until use. The seeds were heated to  $50^{\circ}\text{C}$  for 2 hr, and germinated in an incubator with a constant temperature of  $25^{\circ}\text{C}$  in the dark. Samples were taken from 7:30 a.m. until the root tips of the plants grew to a usable length. Root tips were pretreated with 0.2% colchicine for 40 min before being fixed in Carnoy's solution (glacial acetic acid/absolute ethanol 1:3) for 2 h, then hydrolysed with 1 M hydrochloric acid at a constant temperature of  $60^{\circ}\text{C}$  for 4 min, stained with modified phenol magenta, pressed, and then observed and photographed under a microscope (Olympus BX43).

## 3 | RESULTS

### 3.1 | Genome sequencing and assembly

We sequenced the genome of *Acanthochlamys bracteata* with a nuclear genome size estimated from *k*-mer analysis to be ~245 Mb (Figure S1). Nanopore sequencing yielded 56.35 Gb of raw data and 69.98 Gb of Illumina PCR-free short reads, representing ~235 $\times$  and ~292 $\times$  coverage of the whole genome, respectively. After quality control, 51.34 Gb of clean data was retained, with an average read length of 29.19 kb (Table S3). This draft genome was polished using Illumina short reads and generated a 250-Mb genome assembly with 1161 contigs (contig N50 = 7.55 Mb) and a GC content of 40.3% (Table S4). We removed the redundant contamination found in the genome assembly evaluation (Figure S2, Table S4). The preliminary genome was further assembled using a Hi-C (in vitro fixation of chromosomes) library. The final assembly contained 20 pseudomolecules (Figure 2a,b), which is consistent with the results of a recently published genome assembly (Gao et al., 2021). The final reference assembly was 191.61 Mb in length and included 50 scaffolds (scaffold N50 = 8.62 Mb) and 20 pseudochromosomes. The length of the pseudochromosomes ranged from 7.02 to 14.98 Mb (Table S5). The 20 pseudochromosomes covered 187,720,221 bases or 97.97% of the 191.61-Mb *A. bracteata* assembly, which is higher than that of Gao et al. (2021) (183.84/197.97 Mb: 92.82%).

### 3.2 | Cytological evidence

Previous cytological investigations have shown that *A. bracteata* has a basic chromosome number of  $2n = 2x = 38$  (Kao et al., 1993). However, our results indicate that *A. bracteata* was  $2n = 2x = 40$  (Figure S3), which is consistent with the results of our Hi-C results. Therefore, our work demonstrates that genomic approaches using Hi-C evidence can provide an accurate estimate of the chromosome number of a species.

### 3.3 | Evaluation of assembly quality

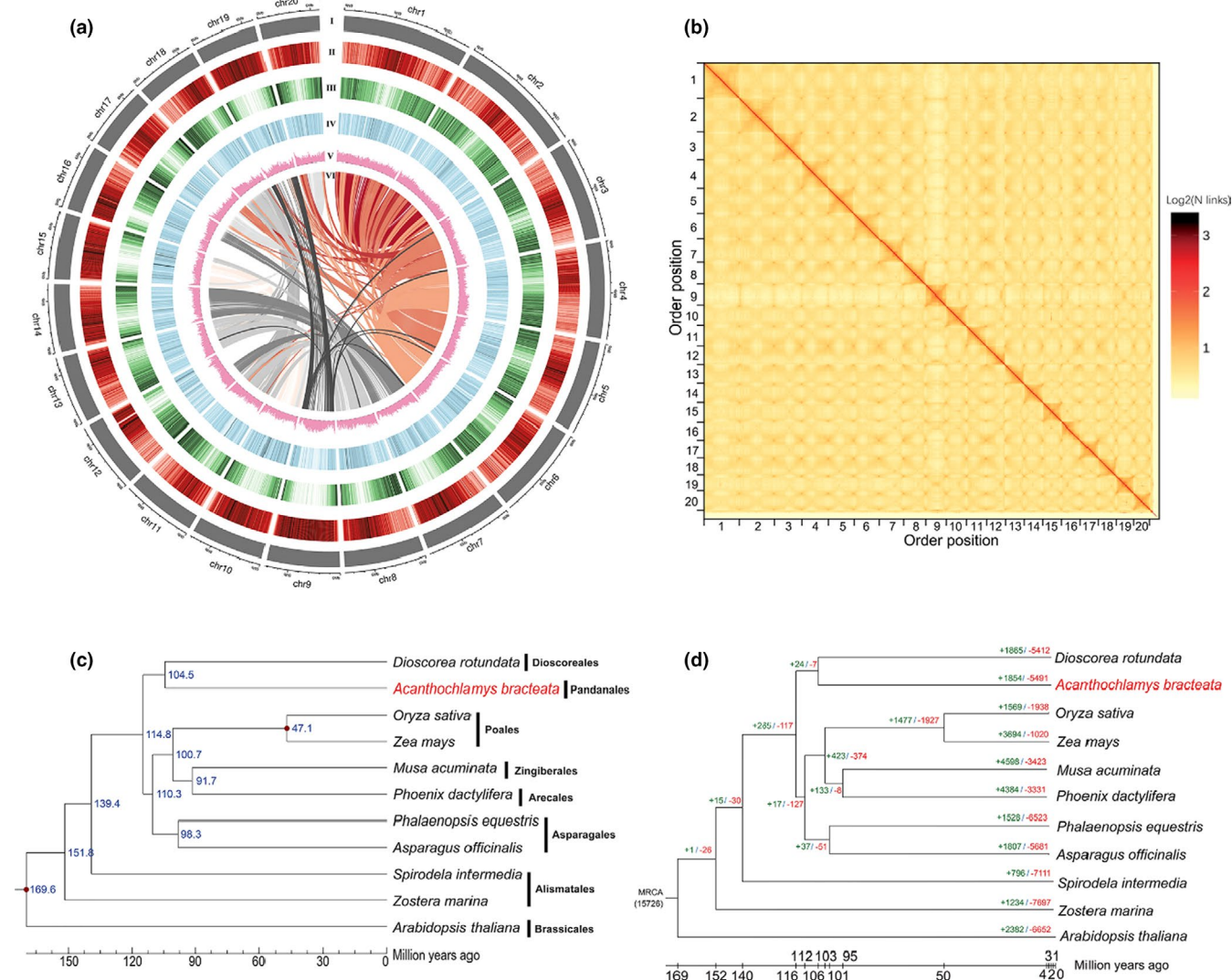
The completeness of our primary assembly and Hi-C assembled genome were evaluated using the BUSCO version 3.1.0 software, with a plant-specific database of 1614 genes. The results indicated that 1,577 (97.7%) genes of the plant single-copy orthologues were complete in both versions of the genome. In the primary assembly, 93.9% ( $n = 1,515$ ) of the genes were complete and in single copies, 3.8% ( $n = 62$ ) were duplicated, 0.7% ( $n = 11$ ) were fragmented and 1.6% ( $n = 25$ ) were missing. The high completeness values were retained in the Hi-C scaffolded assembly, with 94.1% ( $n = 1519$ ) of genes complete and in single copies, 3.6% ( $n = 58$ ) duplicated, 0.6% ( $n = 9$ ) fragmented and 1.7% ( $n = 28$ ) missing. These results indicated that the *A. bracteata* genome assembly was of high quality (Table S6).

To assess the Hi-C assembly quality, we determined the completeness of our primary assembly of *A. bracteata* and the Hi-C assembly (Table 1). Whilst this work was under revision, a genome assembly of *A. bracteata* was published by Gao et al. (2021) and we have also included comparisons with this assembly, which has a slightly lower completeness score (92.8%). A linear comparison of the chromosomes in our assembly and that of Gao et al. (2021) showed good colinearity (Figure S4).

### 3.4 | Repeat annotation, gene prediction and functional annotation

A total of 44.53 Mb of repeat sequences was identified in our assembly, accounting for 17.81% of the genome (Table S7). Among 14.84% transposable elements (TEs) of the genome, 12.71% (31.76 Mb) were predicted to be Class I retrotransposons and 2.13% (5.31 Mb) were predicted to be Class II transposons, which is similar to *Xerophyta viscosa* in which TEs account for 18% of the genome (Costa et al., 2017). In Class I, Copia and Gypsy retrotransposons accounted for 4.28% (10.69 Mb) and 3.94% (9.84 Mb) of the retrotransposons, respectively. In Class II, terminal inverted repeat and Helitron transposons accounted for 1.67% (4.18 Mb) and 0.21% (0.52 Mb) of the transposons, respectively.

De novo prediction, homologous species prediction and Unigene prediction were used to predict protein-coding genes (Table S8). Then, EVIDENCEMODELER was used to integrate the prediction results obtained using the above three methods. Our Venn diagram showed that



**FIGURE 2** Genomic landscape of *Acanthochlamys bracteata* chromosomes and the phylogenetic tree of 11 species. (a) (I) Physical map of 20 assembled chromosomes (Mb scale), numbered according to size. (II) Repeat density represented by proportion of genomic regions covered by repetitive sequences in 100-kb windows. (III) Gene density represented by number of genes in 100-kb windows. (IV) GC content represented by the percentage of G + C bases in 100-kb windows. (V) SNP density represented by the number of SNP markers in 100-kb windows. (VI) Syntenic blocks are depicted by connected lines. (b) Hi-C heatmap. (c) The phylogenetic tree of 11 species; *A. bracteata* and *Dioscorea rotundata* were placed on one branch at ~104.5 Ma. (d) Gene families that contracted and expanded among 11 species: +, number of expanded families; -, number of contracted families. According to the analysis of expansion and contraction based on the constructed phylogenetic tree, a total of 1854 expanded and 5491 contracted families were discovered in *A. bracteata*

21,790 gene models were identified by EVIDENCEMODELER, with an average coding sequence (CDS) length of 1.49 kb (Figure S5). The gene, exon and CDS regions spanned 90.04, 42.97 and 32.53 Mb of the assembly genome, respectively. A total of 56 miRNAs, 1,282 rRNAs and 599 tRNAs were identified (Table S4). In addition, 916 pseudogenes were obtained using GENEWISE. To better understand the gene functions, we searched all 21,294 protein-coding genes against protein databases (Table S8), including NR, KOG, GO, KEGG and TrEMBL. Using these databases, 97.72% of the genes could be annotated. GO terms were classified according to biological processes, cell component and molecular function (Figure S6, Table S9). The distribution of *A. bracteata* genes among the KOG functional classes is shown in Figure S7.

### 3.5 | Comparative genomics and phylogenetic analysis

To investigate the evolutionary relationships between *A. bracteata* and 10 other plant species, we included protein-coding sequences from nine other monocot species: *Asparagus officinalis*, *Dioscorea rotundata*, *Musa acuminata*, *Oryza sativa*, *Phoenix dactylifera*, *Phalaenopsis equestris*, *Spirodela intermedia*, *Zea mays* and *Zostera marina*. The eudicot *Arabidopsis thaliana* was included in the analysis as an outgroup species. A total of 737 single-copy gene families identified across all 11 species were selected to construct an ML phylogenetic tree. The results revealed that *A. bracteata* and



TABLE 1 Comparison of two Hi-C genome assemblies

| Type                                              | HiC genome   | HiC genome<br>(Gao et al., 2021) |
|---------------------------------------------------|--------------|----------------------------------|
| Assembled genome size (Mb)                        | 191.61       | 197.97                           |
| Chromosome anchored length<br>(Mb)/percentage (%) | 187.72/97.97 | 183.84/92.82                     |
| Chromosome number                                 | 2n = 40      | 2n = 40                          |
| GC content (%)                                    | 34.89        | 35.06                            |
| Scaffold N50 (Mb)                                 | 8.62         | 8.64                             |
| Complete BUSCOs (%)                               | 97.7         | 97.0                             |

*D. rotundata* were placed on one branch that diverged ~104.5 Ma (Figure 2c). This placement in the phylogenetic tree was consistent with previous studies (Givnish et al., 2018; Li, Milne, et al., 2020; Li, Wang, et al., 2020) showing that Pandanales and Dioscoreales are sister groups to each other. The phylogenetic relationship of the Zingiberales relative to Poales and Arecales has been unclear (APG III, 2009; Givnish et al., 2010), but our analysis suggested that the Arecales are more closely related to Zingiberales than to Poales. The ancestor of *O. sativa* and *Zea mays* was found to be a sister to *M. acuminata* and *Phoenix dactylifera*, diverging ~100 Ma. The divergence time between *M. acuminata* and *Phoenix dactylifera* was measured as ~91 Ma. We analysed gene family expansion and contraction in 10 monocot species and one eudicot plant species. After diverging from the common ancestor of *A. bracteata* and *D. rotundata*, a total of 1854 and 5491 families were found to have significantly expanded and contracted in *A. bracteata* (Figure 2d), respectively.

### 3.6 | Population structure and phylogenetic analysis

We performed whole-genome resequencing for 110 individuals from 14 populations throughout the species distribution (Figure 1f). From the individuals of each population, we generated an average of 20.48-fold coverage depth, based on the reference genome. After stringent variant calling and filtering, we identified a total of 472,276 high-quality SNPs for further analysis. On the basis of the whole-genome SNPs, PCA, the phylogenetic tree and ADMIXTURE results revealed the first genomic evidence of substantial genetic divergence between the populations.

An analysis of the population structure revealed that the division of the 14 populations into 10 distinct clusters ( $K = 10$ ) reflected geographical divergence (Figures 1f and 3a; Figures S8 and S9). The PCA plot separated the populations west (MK) and east (all except MK) of the Jinsha River, which is consistent with the phylogenetic tree in which population MK is the first to diverge, sister to all the other populations (Figure 4), and also separated the populations north and south of the HDM (Figure 3b). The phylogenetic tree shows that this branch of LH (LH\_RD, LH\_DD) is one of the earlier differentiated populations (Figure 4). Interestingly, the genetic structure of

population LH\_DD split into two clades (Figure 3a). This may be the result of an interchange between glacial and interglacial periods, with contraction during the glacial period and expansion during the interglacial period. Genetic exchange is generated during expansion, which results in the existing genetic structure. Twenty individuals were of mixed ancestry, representing possible hybrids between individuals assigned to different clusters (Figure 3a). Similar conclusions regarding genetic divergence within *A. bracteata* were drawn from an ML tree based on the 472,000 high-quality SNPs (Figure 4), and also from PCA (Figure 3b). In the ML tree, the populations were resolved as robust monophyletic clades, using a bootstrap value of 100%, except for those located in Daofu county (DF\_XS). Of these, one individual (DF\_XS\_4) of this population was nested within another population in the upper Xianshui River, and the other two individuals (DF\_XS\_1 and DF\_XS\_10) clustered with the YJ and LT populations, respectively.

### 3.7 | Demographic history and gene flow

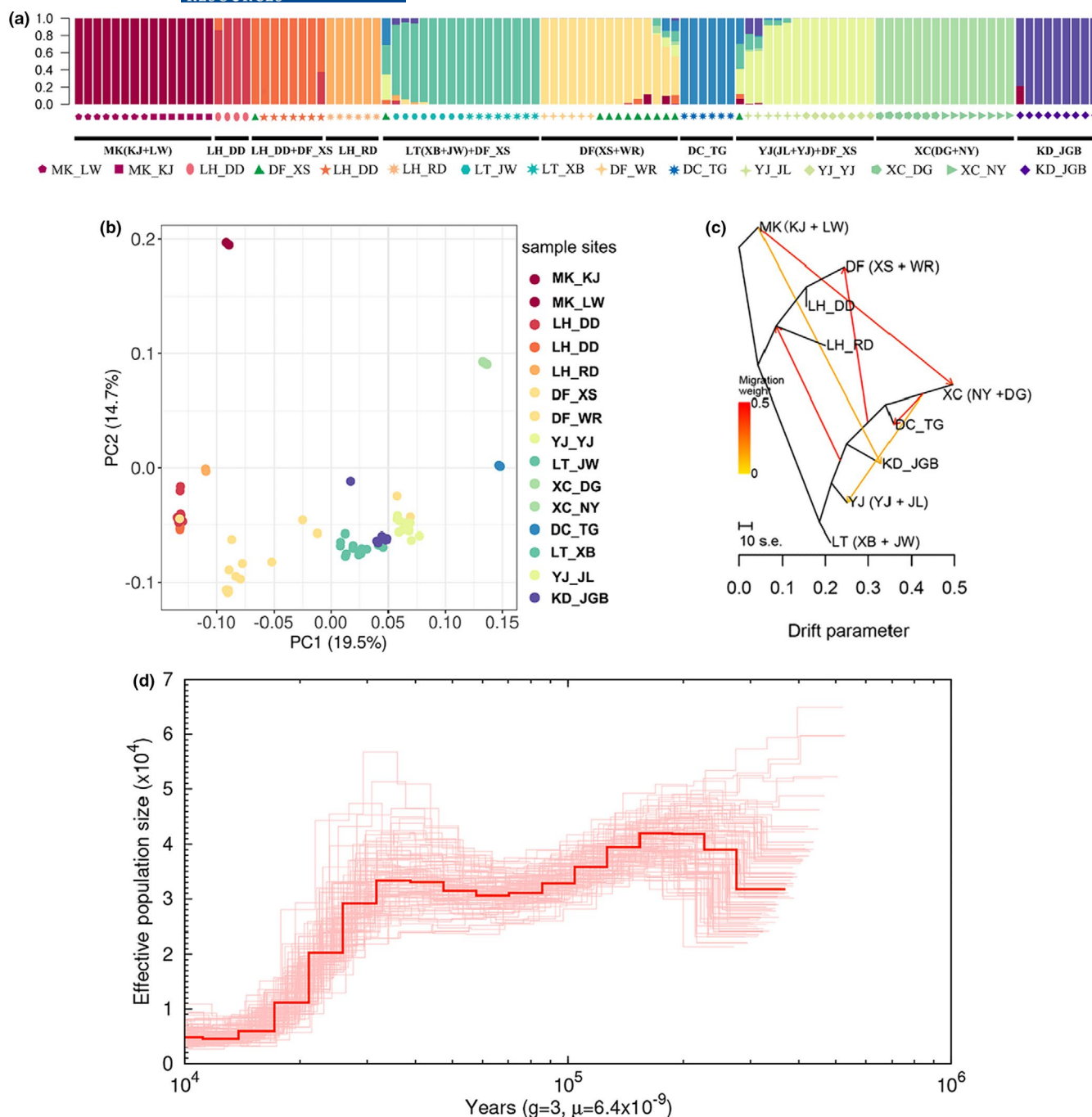
The results of PSMC analysis indicated that the demographic history of *A. bracteata* can be traced back to ~400 ka (Figure 3d; Figure S10). The PSMC plot indicates that the effective population size ( $0.5-4 \times 10^4$ ) of this species has continued to decrease during its evolutionary history. First, the species showed a significant population decline ~70 ka, which coincides with the Baiyu (the Last) Glaciation (Zheng et al., 2002). The Last Glacial (70–10 ka) consisted of two phases, with the middle part of the warm period (30–25 ka) interspersed between periods of cold. The younger phase (the LGM, 25–15 ka) was much colder than the former (Zheng et al., 2002). After the glaciations, the population started to increase very slowly, but the advent of new glaciations resulted in rapid population decline during the LGM (~20 ka).

A population graph (Figure 3c; Figure S11) inferred using TREEMIX shows that the relationships between the populations are similar to those revealed in the ML tree (Figure 4), in whose population history we can see three high levels of gene flow, implying the existence of three migration events. First, as a result of the Baiyu glaciation (~70 ka), the northern taxa (MK) migrated southward (XC) with a gene flow of 0.36. Then, as the glaciers receded, they migrated and spread northward again. The DC and XC populations migrated northward (DF) with a population gene flow of 0.46, and the KD, DC and XC populations migrated northward with a DF and LH population gene flow of 0.40. Finally, the northern populations migrated southward again, due to the LGM, which coincides with a gene flow of 0.41 from the XC population to the DC population.

### 3.8 | Ecological niche modelling

Ecological niche models were constructed for the populations of *A. bracteata* to predict their current and LGM potential distribution. Eight bioclimatic variables (Bio1: annual mean temperature,





**FIGURE 3** Population genetic structure and demographic history. (a) Population structure of 110 *Acanthochlamys bracteata* individuals. The populations grouped by county individual identified are indicated along the x-axis. (b) Principal component analysis (PCA) plot for the 110 *A. bracteata* individuals based on PC1 and PC2. (c) TREEMIX analysis detected significant gene flow from the nine grouped populations. s.e., Standard error. (d) PSMC analysis revealed historical changes in effective population size ( $N_e$ ) for population MK\_LW, with a generation time (g) of 3 years, and a mutation rate ( $\mu$ ) of  $6.4 \times 10^{-9}$  per site per generation. The time axis is log-transformed

Bio2: mean diurnal range, Bio3: isothermality, Bio4: temperature seasonality, Bio12: annual precipitation, Bio14: precipitation of the driest month, Bio18: precipitation of the warmest quarter, Bio19: precipitation of coldest quarter) were retained with correlation coefficients  $<0.7$ . The AUC value of the model was 0.998 for this species. Our model showed that the distribution range during the LGM

was narrower than the current potential distribution (Figure 5a,b; Figure S12). With the start of the LGM, the distribution of this species showed a clear trend toward constriction, due to the changes in the climate and environment, and once the glaciation receded and the temperature rebounded, the distribution range expanded significantly.

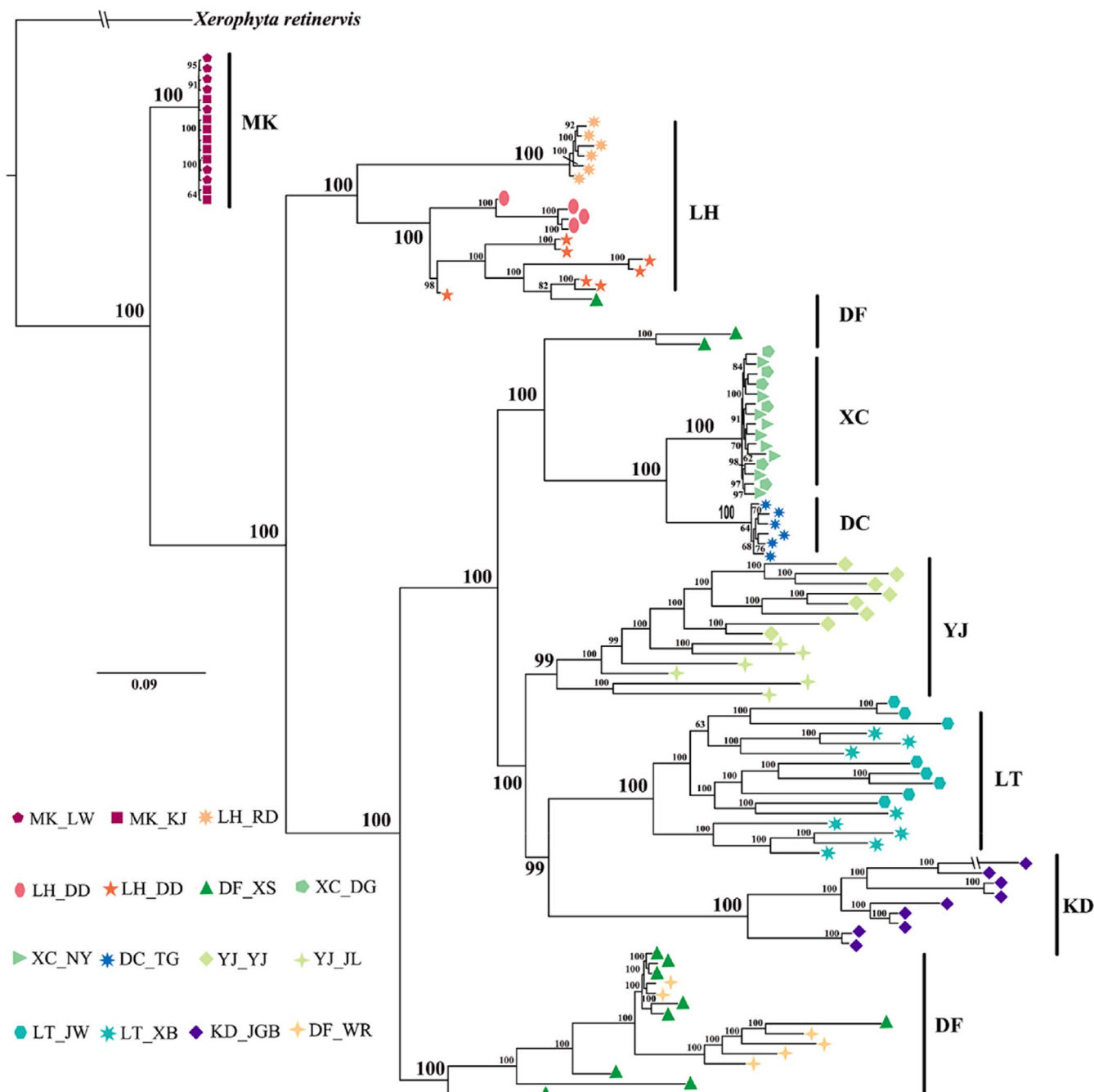


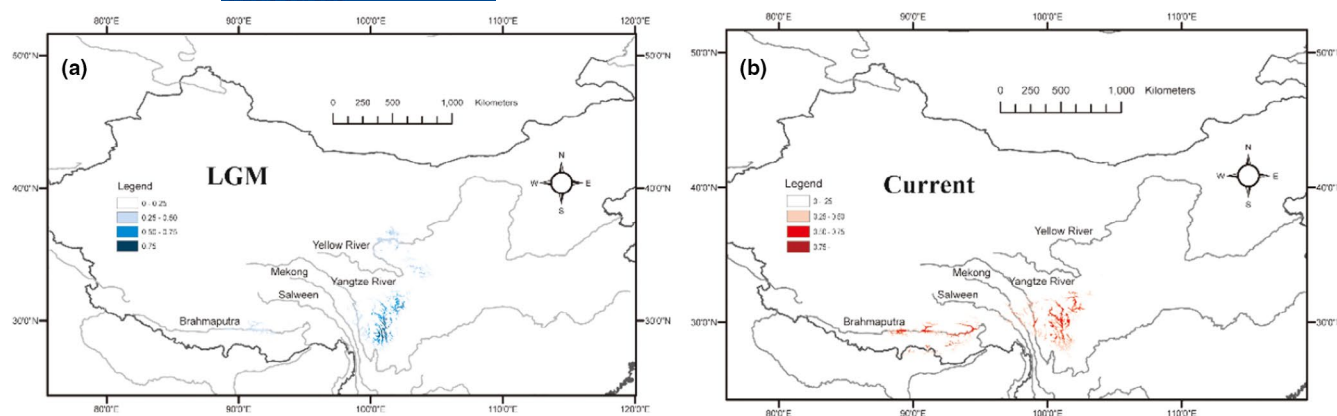
FIGURE 4 Maximum-likelihood tree based on SNPs from whole-genome resequencing. The populations grouped by county individual identified are indicated along the y-axis

## 4 | DISCUSSION

### 4.1 | Population structure, demographic history and gene flow among *Acanthochlamys bracteata* lineages

Our findings indicated that Quaternary climatic changes played an important role in shaping the genetic structure and demographic trajectories of this species, and provided critical insights into the genetic status and evolutionary history of this poorly understood

species. The geographical structure of the 10 *A. bracteata* lineages detected here implies that the Jinsha River and the mountains on the east bank of the Jinsha River may have limited gene flow between populations. However, our whole-genome resequencing based on population genetic analysis revealed that three different genetic exchanges occurred sequentially between the isolated populations of the HDM, a finding which is consistent with the transitions between ice ages and interglacial periods (Zheng et al., 2002). From the structure (Figure 3a) and gene flow (Figure 3c) plots, it can be seen that locations with higher extents of gene exchange show a



**FIGURE 5** Ecological niche models for *Acanthochlamys bracteata*. (a) LGM potential distribution of the species predicted by MAXENT. (b) Current potential distributions of the species predicted by MAXENT

tendency to heterozygosity in structure, according to ML tree analysis. Therefore, the occurrence of multiple migratory events is also indicative of frequent gene exchange along the east bank of the Jinsha River with Quaternary climate change.

## 4.2 | Conservation implications

The overall genetic diversity of this species is low, due to the population decline caused by the glaciations, and the changes in environmental climate after the formation of the Jinsha River, following uplift of the Qinghai–Tibet Plateau. Conservation measures must therefore be designed taking these details into account. We suggest separating the 14 populations into two groups based on nucleotide diversity ( $\theta_\pi$ ,  $\theta_w$ , Table S2): one group with lower genetic diversity (MK\_KJ, MK\_LW, LH\_RD, DC\_TG, XC\_DG and XC\_NY) and a second group with higher genetic diversity (YJ\_YJ, YJ\_JL, LT\_JW, LT\_XB, KD\_JGB, LH\_DD, DF\_XS and DF\_WR). The two groups face very different conservation issues, so we recommend that each group should be assessed independently, according to its genetic diversity status.

Our preliminary survey found that areas with low genetic diversity of *A. bracteata* tend to be located in alpine meadows with overall low species diversity. For example, the two populations MK\_KJ and MK\_LW were located in Rumei county near the Lancang River (Figure 1). However, this river is highly suitable for hydropower development, and if this occurs large areas of potential habitat of the MK\_KJ and MK\_LW populations could be flooded and occupied. Due to the lagging local economy, the development of roads and animal husbandry has caused serious damage to the population. Hence, the MK population faces a high risk of extinction, and seed collection from populations threatened by development, for *ex situ* conservation, is necessary.

Areas with high genetic diversity should be targeted for priority conservation, because the conservation of biogenetic diversity implies the conservation of species diversity (Oliver et al., 2015). The conservation of these areas should focus on protecting habitats and maintaining regional environmental stability to avoid destruction,

with specific protection measures to establish local conservation plots. Second, we recommend the establishment of long-term population dynamics monitoring institutions at the local level, to facilitate artificial protection and population recovery.

From a broader perspective, our study emphasizes the value of genome evidence in conservation genetics for threatened species that have a complex genetic structure and evolutionary history.

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## CONFLICTS OF INTEREST

The authors declare no competing financial interests.

## AUTHOR CONTRIBUTIONS

B.X. and X.G. designed the research. B.X., H.D., M.L., W.J., L.J. and J.Z. collected the samples and performed the research. B.X., C.Y., Y.G., X.L. and Y.L. analysed the data. B.X. and M.L. wrote the paper. All authors read and approved the final manuscript.

## DATA AVAILABILITY STATEMENT

All the data sets (Illumina, Nanopore, Hi-C and assembly files) are deposited in the China National Center for Bioinformation-National Genomics Data Center (<https://bigd.big.ac.cn/>) website with BioProject accession no. PRJCA004826. Additional information may be found the Supporting Information. Genome sequences have been submitted to the National Genomics Data Center (NGDC). Nanopore whole-genome sequencing data, Illumina data, transcriptome data and resequencing data have

been deposited in BioProject/GSA (<https://bigd.big.ac.cn/gsa/>) under accession number project PRJCA004826 and the whole-genome assembly and annotation data have been deposited in BioProject/GWH (<https://ngdc.cncb.ac.cn/gwh/>) under accession No. GWHBAYO00000000.1 and Hi-C genome assembly was under accession No. GWHBAYO00000000.

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